Université de Haute-Alsace
Faculté des Sciences et Techniques

M2 – IM - IMDS - MIAGE

Statistical and Machine Learning

HW2 : Uniform Convergence and the Limits of Learning

Expected Deliverables

Your submission should consist of a single Jupyter notebook (`.ipynb`) containing:

- well-structured and documented code;
- appropriate visualizations;
- comments interpreting each experiment;
- comparisons between experimental observations and the theoretical quantities derived in this homework;
- a short conclusion summarizing what your experiments reveal about uniform convergence, the No-Free-Lunch principle, and inductive bias.

The completed notebook must be submitted by uploading it to the course GitHub repository. Unless otherwise specified, the submission deadline is **one week after the HW is distributed**.

Objective

The objective of this homework is to study **uniform convergence** and the **limits of learning** through repeated probabilistic simulations.

Unlike the TD session, the goal here is not to manipulate specific geometric hypothesis classes by hand. Instead, you will directly simulate the probability distributions that appear in the theoretical results.

You will first study how empirical risks fluctuate around true risks and how this behavior depends on the sample size and the size of the hypothesis class. You will then simulate learning problems with completely random target functions to illustrate the No-Free-Lunch phenomenon. Finally, you will introduce structure into the target functions and observe how an appropriate inductive bias enables generalization.

You are expected to choose appropriate values for experimental parameters such as:

- sample sizes m;
- hypothesis-class sizes $|\mathcal{H}|$;
- accuracy levels ϵ;
- domain sizes N;
- numbers of Monte-Carlo repetitions.

Your choices should be justified and should be sufficiently varied to support your conclusions.

No machine learning libraries (e.g., `scikit-learn`) are required. You may use standard numerical and visualization libraries such as `numpy` and `matplotlib`.

Part I – Simulating Uniform Convergence

Consider a finite hypothesis class

$$\mathcal{H} = \{h_1, \ldots, h_K\}.$$

Suppose that each hypothesis h_j has a fixed true error

$$L_D(h_j) = p_j, \qquad p_j \in [0, 1].$$

For a training sample of size m, let E_j denote the number of errors made by h_j on the training sample.

In this simplified simulation, assume

$$E_j \sim \text{Binomial}(m, p_j),$$

so that the empirical risk is

$$L_S(h_j) = \frac{E_j}{m}.$$

This allows us to study the fluctuations of $L_S(h)$ around $L_D(h)$ directly.

Tasks

1. **Empirical Risk as a Random Variable**

   Choose several values

   $$p_1, \ldots, p_K$$

   for the true risks of the hypotheses.

   For a fixed hypothesis h_j and several sample sizes m:

   (a) simulate $E_j \sim \text{Binomial}(m, p_j)$ many times;

   (b) compute

   $$L_S(h_j) = \frac{E_j}{m};$$

   (c) visualize the distribution of $L_S(h_j)$;

   (d) compare its empirical mean with the true risk

   $$L_D(h_j) = p_j.$$

   Discuss how the variability of $L_S(h_j)$ changes as m increases.

2. **Deviation Between Empirical and True Risk**

   For every simulated training sample, compute

   $$|L_S(h_j) - L_D(h_j)|.$$

   For a chosen value of ϵ, estimate

   $$\Pr\left[|L_S(h_j) - L_D(h_j)| > \epsilon\right].$$

   Repeat the experiment for several values of m.

   Compare the experimental probability with the Hoeffding bound

   $$\Pr\left[|L_S(h_j) - L_D(h_j)| > \epsilon\right] \leq 2e^{-2m\epsilon^2}.$$

   Plot both quantities as functions of m.

   Discuss whether the theoretical bound is exact or conservative.

3. **From One Hypothesis to Uniform Convergence**

   We now consider all hypotheses in $\mathcal{H}$ simultaneously.

   For every simulated sample, generate empirical risks

   $$L_S(h_1), \ldots, L_S(h_K)$$

   and compute

   $$\Delta_S = \max_{h \in \mathcal{H}} |L_S(h) - L_D(h)|.$$

   Estimate experimentally

   $$\Pr[\Delta_S > \epsilon].$$

   Compare your result with the finite-class uniform-convergence bound

   $$\Pr\left[\sup_{h \in \mathcal{H}} |L_S(h) - L_D(h)| > \epsilon\right] \leq 2|\mathcal{H}|e^{-2m\epsilon^2}.$$

   Repeat the experiment for several sample sizes.

4. **Effect of the Size of the Hypothesis Class**

   Fix ϵ and repeat the uniform-convergence experiment using several values of

   $$|\mathcal{H}|.$$

   For example, you may consider hypothesis classes containing approximately

   $$5, \quad 10, \quad 50, \quad 100, \quad 500$$

   hypotheses.

   Plot

   $$\Pr[\Delta_S > \epsilon]$$

   as a function of m for the different class sizes.

   Discuss the following questions:

   - How does increasing $|\mathcal{H}|$ affect uniform convergence?
   - Does a larger hypothesis class generally require more data?

- How does this relate to the $\log |\mathcal{H}|$ dependence appearing in finite-class sample-complexity bounds?

5. **Experimental Sample Complexity**

   Fix values of ϵ and δ.

   Experimentally determine the smallest tested value of m for which

   $$\Pr[\Delta_S > \epsilon] \leq \delta.$$

   Compare this experimental value with the theoretical sufficient sample size obtained from

   $$m \geq \frac{\log(2|\mathcal{H}|/\delta)}{2\epsilon^2}.$$

   Discuss why the theoretical bound should be interpreted as a **sufficient guarantee** rather than necessarily the exact minimum sample size.

Part II – Simulating No-Free-Lunch

Consider now a finite input domain

$$\mathcal{X} = \{1, \ldots, N\},$$

with the uniform distribution

$$\Pr(X = x) = \frac{1}{N}, \qquad x \in \mathcal{X}.$$

For each experimental repetition, generate a new target function

$$f : \mathcal{X} \to \{0, 1\}$$

by independently sampling

$$f(x) \sim \text{Bernoulli}\left(\frac{1}{2}\right)$$

for every $x \in \mathcal{X}$.

Each repetition therefore corresponds to a new randomly generated learning task.

The learner receives a training sample

$$S \sim D^m.$$

Use the following simple learning algorithm:

- if an input x has appeared in the training sample, predict its observed label;
- if x has never appeared in the training sample, predict 0.

6. **Seen and Unseen Inputs**

   For several choices of N and m:

(a) generate a random target function f;

(b) generate a training sample of size m;

(c) compute the fraction of inputs in $\mathcal{X}$ that were never observed in the sample.

Repeat the experiment many times and estimate the expected fraction of unseen inputs.

Compare your simulation with the theoretical probability that a fixed point $x \in \mathcal{X}$ is never sampled:

$$\Pr[x \text{ is unseen}] = \left(1 - \frac{1}{N}\right)^m.$$

Explain why this is also the expected fraction of unseen points.

7. **Error on Unseen Inputs**

   Evaluate the true risk of the memorization learner over the entire domain:

$$L_D(A(S)) = \frac{1}{N} \sum_{x \in \mathcal{X}} \mathbf{1}[A(S)(x) \neq f(x)].$$

   Repeat the experiment over many randomly generated target functions and training samples.

   Since the labels of unseen inputs are independent Bernoulli(1/2) variables, derive the theoretical prediction

$$\mathbb{E}[L_D(A(S))] = \frac{1}{2}\left(1 - \frac{1}{N}\right)^m.$$

   Compare this formula with your simulations.

   Plot the theoretical and experimental expected risks as functions of m.

8. **Changing the Prediction on Unseen Inputs**

   Modify the learner so that unseen inputs are handled using different rules, for example:

   - always predict 0;
   - always predict 1;
   - predict a random Bernoulli(1/2) label.

   Compare their expected true risks when target functions are sampled completely at random.

   Do any of these strategies consistently outperform the others?

   Explain how your observations relate to the No-Free-Lunch principle.

Part III – Introducing Prior Knowledge

We now modify the previous experiment.

Keep the same finite input domain

$$\mathcal{X} = \{1, \ldots, N\},$$

but instead of generating an arbitrary random target function, generate a structured threshold target

$$f_\theta(x) = \mathbf{1}[x > \theta],$$

where

$$\theta \in \{1, \ldots, N-1\}.$$

The target function is therefore no longer an arbitrary labeling of the domain.

Consider a learner whose hypothesis class consists of threshold functions:

$$\mathcal{H} = \{h_t(x) = \mathbf{1}[x > t] : t \in \{1, \ldots, N-1\}\}.$$

Use ERM to select a threshold from the training sample.

9. **Learning with a Matching Inductive Bias**

   Implement ERM over the threshold hypothesis class.

   For several values of m:

   (a) generate a threshold target f_θ;
   (b) draw a training sample;
   (c) learn h_S using ERM;
   (d) compute the true risk over the complete finite domain $\mathcal{X}$.

   Repeat the experiment many times and estimate

   $$\mathbb{E}[L_D(h_S)].$$

   Compare these results with the random-target experiment from Part II.

10. **Random Targets vs. Structured Targets**

    Produce a visualization comparing the evolution of the true risk with m for:

    (a) completely random target functions;
    (b) structured threshold target functions learned using a threshold hypothesis class.

    Discuss why the learner can generalize to unseen inputs much more effectively in the second setting.

    In particular, explain the role played by the assumption that the target belongs to the threshold family.

11. **Final Analysis**

    Write a short conclusion connecting all experiments.

    Your discussion should address the following points:

    - why $L_S(h)$ becomes more concentrated around $L_D(h)$ as m increases;
    - why uniform convergence requires simultaneous control over all hypotheses;
    - how increasing $|\mathcal{H}|$ affects this simultaneous control;
    - why finite-class theoretical bounds can be conservative;
    - why completely random targets make unseen labels unpredictable;
    - how the No-Free-Lunch experiment demonstrates that a learner cannot infer arbitrary unseen labels from the training sample alone;
    - how restricting the target and hypothesis classes to thresholds introduces prior knowledge;
    - why this inductive bias allows the learner to generalize beyond the observed examples.